

Industrial IoT Testing Patterns

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Scope: General Knowledge Base

Intended use: reusable test patterns for gateways, devices, telemetry, and edge systems

Owner: Industrial Systems QA

1. Purpose

This document provides reusable testing patterns for industrial IoT systems, including gateways, protocol adapters, telemetry pipelines, cloud forwarding, local buffering, device management, and operational monitoring.

These patterns can be reused when generating test plans for industrial gateways, connected machines, sensor networks, and edge-computing systems.

2. Typical Industrial IoT Architecture

An industrial IoT system often contains:

- devices, machines, PLCs, sensors, or controllers;
- protocol adapters;
- edge gateway;
- local database or buffer;
- cloud ingestion API;
- monitoring dashboard;
- identity and access control;
- remote configuration service;
- audit and diagnostics service.

Testing should cover both local continuity and cloud integration.

3. Core Risk Areas

3.1 Data Loss

Telemetry may be lost during:

- network outage;
- gateway restart;
- buffer overflow;
- adapter crash;
- malformed protocol payload;
- cloud ingestion timeout.

3.2 Duplicate Data

Duplicate telemetry may occur when:

- a retry is executed after partial success;
- event IDs are not stable;
- cloud acknowledgement is ambiguous;
- buffered batches are replayed incorrectly.

3.3 Timing and Ordering

Industrial systems often require:

- correct timestamp handling;
- chronological replay;

- event ordering;
- latency constraints;
- polling interval enforcement.

3.4 Configuration Risk

Bad configuration can cause:

- wrong signal mapping;
- wrong unit conversion;
- excessive polling;
- wrong protocol endpoint;
- security exposure;
- device disconnect.

3.5 Security and Remote Access

Remote management can introduce:

- unauthorized configuration changes;
- exposed diagnostic bundles;
- weak certificate handling;
- unsafe support access.

4. Test Pattern: Device Registration

Objective:

- verify that devices can be registered, validated, updated, disabled, and audited.

Test ideas:

- register valid device;
- register duplicate device ID;
- register invalid IP or hostname;
- register unsupported protocol;
- update signal map;
- disable device;
- verify audit event for each change.

Evidence:

- API response;
- device inventory;
- audit log;
- adapter runtime status.

5. Test Pattern: Protocol Adapter Validation

Objective:

- verify protocol-specific behavior.

For Modbus TCP:

- valid register read;
- invalid register address;
- timeout;
- wrong unit conversion;
- reconnect after disconnect.

For OPC UA:

- valid node subscription;
- invalid node ID;
- certificate error;
- session timeout;
- reconnect behavior.

For MQTT:

- topic subscription;
- malformed payload;
- retained message handling;
- broker disconnect;
- duplicate message ID.

For HTTP sensor adapter:

- valid sensor response;
- HTTP 500;
- timeout;
- invalid JSON;
- authentication failure.

6. Test Pattern: Telemetry Normalization

Objective:

- verify raw protocol data is converted into a common schema.

Check:

- device ID;
- signal ID;
- original timestamp;
- ingestion timestamp;
- value;
- unit;
- quality flag;
- source protocol;
- event ID.

Negative tests:

- missing timestamp;
- unsupported unit;
- non-numeric value where numeric expected;
- future timestamp;
- stale timestamp;
- unknown signal ID.

7. Test Pattern: Local Buffering

Objective:

- verify telemetry is preserved when cloud is unavailable.

Scenario:

1. Start telemetry flow.
2. Disable cloud ingestion.
3. Continue telemetry for defined duration.
4. Verify records are buffered.
5. Restore cloud ingestion.

6. Verify chronological replay.
7. Verify no duplicates.

Metrics:

- records generated;
- records buffered;
- records forwarded;
- records dropped;
- retry count;
- buffer capacity.

8. Test Pattern: Cloud Forwarding Retry

Objective:

- verify retry behavior under transient cloud failures.

Test cases:

- HTTP 429 rate limit;
- HTTP 500;
- TCP timeout;
- partial batch acceptance;
- invalid TLS certificate;
- cloud unavailable for multiple retry cycles.

Expected results:

- exponential backoff is applied;
- retry count is visible;
- alert threshold is enforced;
- local collection continues;
- no duplicate records are accepted.

9. Test Pattern: Device Health State

Objective:

- verify health transitions are accurate and timely.

States:

- healthy;
- degraded;
- disconnected;
- configuration_error.

Test triggers:

- adapter timeout;
- invalid signal mapping;
- protocol authentication failure;
- repeated adapter crash;
- device reconnect.

Evidence:

- health API response;
- UI status;
- adapter logs;
- alert event.

10. Test Pattern: Time Synchronization

Objective:

- verify timestamp trustworthiness.

Tests:

- NTP available at startup;
- NTP unavailable at startup;
- NTP lost during operation;
- local clock drift;
- timestamp flagged as unsynchronized;
- replay uses original timestamp and ingestion timestamp correctly.

11. Test Pattern: Configuration Versioning

Objective:

- verify every configuration change is versioned and reversible.

Tests:

- create initial configuration;
- update signal map;
- compare current and previous versions;
- apply invalid configuration;
- roll back to previous valid version;
- verify audit history remains complete.

12. Test Pattern: Diagnostics

Objective:

- verify diagnostic bundles support troubleshooting without exposing secrets.

Bundle should include:

- system info;
- active configuration;
- adapter status;
- buffer statistics;
- recent logs;
- audit events.

Bundle should exclude or redact:

- passwords;
- private keys;
- API tokens;
- bearer tokens;
- cloud credentials.

13. Test Pattern: Long-Run Operation

Objective:

- verify stability under sustained load.

Long-run tests should monitor:

- memory usage;
- CPU usage;

- disk growth;
- buffer size;
- adapter restarts;
- cloud forwarding lag;
- data loss;
- duplicate records.

Compressed endurance profiles may be used when a full-duration test is not practical, but the compression assumptions should be documented.

14. Recommended Test Data

Use data that includes:

- normal operating values;
- boundary values;
- missing values;
- invalid units;
- stale timestamps;
- future timestamps;
- duplicate event IDs;
- high-frequency bursts;
- device disconnects;
- malformed payloads.

15. Demo Questions for Knowledge Retrieval

The knowledge base should be able to answer:

- How should we test local buffering during a cloud outage?
- What evidence should be collected for diagnostic bundle tests?
- Which risks are common in industrial IoT telemetry systems?
- How do we test duplicate telemetry prevention?
- What should be checked for protocol adapter validation?

16. References

- General systems testing practice from ISO/IEC/IEEE 29119-style test processes.
- Security assessment concepts from NIST SP 800-115.
- Verification and traceability concepts from NASA systems-engineering practice.